SESSION 6 – LAB REPORT

Dice Roller Console Program

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1. Aim

To develop a Kotlin console program that simulates rolling a pair of dice using random number generation. The program demonstrates the use of functions, state updates, and conditional logic by displaying a special message when both dice show the same number.

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2. Problem Statement

Build a console-based Dice Roller program in Kotlin that simulates pressing a roll button three times. Each roll should generate two random numbers between 1 and 6. The program must display the result of each roll and print a special message if both dice show the same value (double).

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3. Program Code

```
import kotlin.random.Random

// Function to roll a single die fun
rollDie(sides: Int, rnd: Random): Int {
return rnd.nextInt(1, sides + 1)
}

// Function to roll two six-sided dice fun
rollTwoDice(rnd: Random): Pair<Int, Int> {
val firstDie = rollDie(6, rnd)     val
secondDie = rollDie(6, rnd)     return
Pair(firstDie, secondDie)
}

// Function to format roll output
fun formatRoll(rollNumber: Int, a: Int, b: Int): String {
var result = "Roll $rollNumber: $a and $b"
```

```
        if (a ==
b) {
result += " ->
Double!"
    }
    return result
}  fun
main() {
    println("--- Dice Roller ---")

    val rnd = Random(123)  // Seeded for reproducible output
var rollNumber = 0

    repeat(3) {
        println("Pressing 'Roll'...")
rollNumber++
         val (die1, die2) = rollTwoDice(rnd)
println(formatRoll(rollNumber, die1, die2))
    }
                                                                                    }
```

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4. Sample Output

```
--- Dice Roller ---
Pressing 'Roll'...
Roll 1: 2 and 5
Pressing 'Roll'...
Roll 2: 6 and 6 -> Double!
Pressing 'Roll'...
                                                                    Roll 3: 1 and 4
```

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5. Explanation

This program uses Kotlin’s Random class to simulate rolling dice. The rollDie() function generates a random number between 1 and the given number of sides. The rollTwoDice() function calls rollDie() twice to simulate two dice and returns the result as a Pair. The formatRoll() function formats the output and checks whether both dice values are equal. If they are equal, it appends a special message indicating a double. The main() function simulates pressing the roll button three times and maintains state using the rollNumber variable.

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6. Concepts Used

- Kotlin Functions
- Random Number Generation
- Conditional Statements (if condition)
- String Templates

- Pair Data Structure
- State Management using Variables

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7. Conclusion

This lab helped in understanding how state changes after each simulated button press. The program demonstrates how user interaction leads to state updates and output generation. It also strengthened knowledge of functions, randomness, and conditional logic in Kotlin.